

Off-anchor robustness of the A-zero uniqueness inequalities and the load-bearing constants (gate ROBUSTNESS-MU2, partial advance)

TECT verification-first repository · claim B1-RH-ENUM

first issued 2026-06-06 · this version issued 2026-06-06 · v1.0

1. Purpose and scope

The whole Sector-B analysis is pinned to the production anchor $\mu^2 = 0.005$; ROBUSTNESS-MU2 asks whether the Reading-H selection survives in a μ^2 -neighbourhood. This note advances the gate by settling its A=0-uniqueness component and bounding the μ^2 -variation of the constants the remaining components depend on. It does NOT close the gate: the exact off-anchor re-margining of the STEP-5B closure is registered as the narrowed residual.

2. Structural μ^2 -independence of the sign decomposition

The lemmas behind the A=0 uniqueness (note ga0-dui-closure) use only: $u < 0$, $v > 0$, $M' < 0$ (L1, μ^2 -independent), $u_{\text{eff}}(M_c) = u + 10vM_c = 0$ with $M_c = -u/(10v)$ (μ^2 -independent), and the closed-form bound of L3. NONE references the specific value of μ^2 . Hence the sign decomposition holds for every $\mu^2 > 0$ by structure; only the two anchor inequalities $m^* > m_w$ and $M_R > M_c$ carry the anchor value.

3. The μ^2 -cancellation and the robustness radius

Proposition 1 (μ^2 cancels in the margin). $m^* - m_w = 3u M_R + 15v (M_R^2 - M_c^2)$, independent of μ^2 except through $M_R(\mu^2)$.

Proof. $m^* = \mu^2 + 3uM_R + 15vM_R^2$ (gap equation) and $m_w = \mu^2 + 15vM_c^2$; subtracting removes μ^2 . (Verified to 10^{-6} at $\mu^2 = 0.0013, 0.005, 0.02$.) $\square$

Since $M_R(\mu^2)$ is monotone and varies by $< 2\%$ across the $\times 4$ band ($0.1097 \rightarrow 0.1084$), the margin $m^* - m_w$ is nearly flat. Numerically (sweep $\mu^2 \in [0.00125, 0.02]$, $\times 0.25$ to $\times 4$): m^*/m_w ranges $8.37 \rightarrow 5.74$ and M_R/M_c ranges $4.13 \rightarrow 4.08$ – both comfortably above 1 throughout. Extending the fine sweep, $m^*/m_w > 2$ and $M_R/M_c > 1.5$ hold on $\mu^2 \in [0.001, 0.05] = [\times 0.2, \times 10]$ of the anchor. The A=0 uniqueness is therefore robust on an order-of-magnitude neighbourhood.

4. Constants for the remaining components

The layer margin (Prop A) and the STEP-5B closure margins depend on μ^2 through the dressed constants $\lambda'(\mu^2) = 3u + 30vM_R$ and $\hat{r}(\mu^2)$. Across the $\times 4$ band λ' decreases smoothly and mono-

tonically $8.080 \rightarrow 7.957$ (a 1.5% variation, no sign change) and M_R by $< 2\%$. A variation this small cannot collapse the STEP-5B floors ($\times 59.4 / \times 8.8 / \times 2.6$ at the anchor) or the layer margin; however, the EXACT off-anchor margins are not recomputed here (the full beyond-layer machinery is one anchor-intensity per run). This is the registered residual: *re-margin the STEP-5B closure on the μ^2 -neighbourhood*.

5. Numerical verification

Script `codes/vacuum/robustness_mu2_sweep.py` (v1.0.0; 9/9 self-test asserts PASS; artefact `claims/B1-RH-ENUM/runs/260606-robustness-mu2/result.json`).

Quantity	Value	Check
$M_R > M_c$ on $\times 4$ band	min ratio 4.08	holds throughout
$m^* > m_w$ on $\times 4$ band	min ratio 5.74	holds throughout
robustness radius ($m^*/m_w > 2$, $M_R/M_c > 1.5$)	$\mu^2 \in [0.001, 0.05]$	$\times 0.2$ to $\times 10$
μ^2 -cancellation identity	exact to 10^{-6}	margin = $3uM_R + 15v(M_R^2 - M_c^2)$
λ' across $\times 4$	$8.080 \rightarrow 7.957$ (1.5%)	smooth, monotone, > 0
anchor reproduces certified	M_R, m^* to 5×10^{-5}	consistency

Quantitative sanity check (mandatory): at the anchor the sweep reproduces the certified $M_R = 0.109414$ and $m^* = 0.304526$; the margin $m^*/m_w = 7.76$ matches the G-A0-VER value, confirming the sweep shares the certified calibration.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3.5)

- (α) “*You verified the $A=0$ inequalities but not the actual STEP-5B closure off-anchor – calling ROBUSTNESS-MU2 ‘advanced’ overstates.*” VALID – and explicitly mitigated: the gate is NOT closed. The $A=0$ -uniqueness component is settled (structural + $\times 0.2.. \times 10$ verification); the STEP-5B/layer re-margin is the registered residual. Only that scope is claimed.
- (β) “*Is the $\times 10$ range physically meaningful?*” VALID-with-mitigation: the anchor $\mu^2 = r_{\text{braz}} = 0.005$ sits at the Brazovskii operating point; $[\times 0.2, \times 10]$ covers the physically relevant range comfortably. The radius is measured and reported, not assumed.
- (γ) “*A 1.5% λ' variation – does it bound the margin variation?*” VALID-with-mitigation: a 1.5% constant shift cannot collapse a $\times 2.6$ floor, but the exact margin sensitivity is the registered re-margin sub-task; the claim here is ‘numerically supported’, not ‘proven’.
- (δ) “*Could $M' < 0$ fail off-anchor?*” DISMISSED: L1’s dominated-convergence proof (G-A0-DUI) holds for every $m > 0$ with no μ^2 reference; $M' < 0$ is μ^2 -independent.

No objection upheld as a blocker; the honest scope is a PARTIAL advance.

7. Result footer

Result ID: B1-RH-ENUM / ROBUSTNESS-MU2 partial advance
Precise statement: Sign-decomposition lemmas are μ^2 -independent;
the anchor inequalities $m^* > m_w$, $M_R > M_c$ hold on

	μ^2 in [0.001,0.05] ($\times 0.2.. \times 10$), min ratios 5.74/4.08 across x_4 ; $m^*-m_w = 3uM_R+15v(M_R^2-M_c^2)$ (μ^2 cancels); λ' varies 1.5% across x_4 . A=0 uniqueness robust; STEP-5B/layer off- anchor re-margin = narrowed residual.
Scope:	μ^2 -neighbourhood of 0.005. A=0-uniqueness component settled; STEP-5B/layer component numerically supported, not re-margined.
Dependencies:	ga0-dui-closure (L1/L2/L3); A1-KERNEL-CONV.
Evidence grade:	ANALYTIC (μ^2 -independence + cancellation) + EXECUTED (9/9 sweep)
Reproduction command:	python codes/vacuum/robustness_mu2_sweep.py
Expected output:	claims 9/9 PASS; ratios table; result.json under claims/B1-RH-ENUM/runs/260606-robustness-mu2/
Falsification gate:	A μ^2 in [0.001,0.05] with $m^* \leq m_w$ or $M_R \leq M_c$ (refutes the radius); a STEP-5B margin going below 1 off-anchor (would reopen fully).
Tier before / after:	ROBUSTNESS-MU2 stays OPEN (narrowed): A=0- uniqueness component robust; STEP-5B re-margin residual. No tier-number change on B1/B2.
No-overclaim statement:	The gate is NOT closed. Only the A=0-uniqueness robustness + constant-smoothness evidence is claimed; the exact off-anchor STEP-5B margins are numerically supported (1.5% constant drift), not recomputed. No tier-number change.
Next required action:	Re-margin the STEP-5B closure on the μ^2 - neighbourhood (run beyond_layer_gershgorin_bound at off-anchor μ^2) to fully close ROBUSTNESS-MU2.